Environmental Modelling & Software 39 (2013) 159–175

Contents lists available at SciVerse ScienceDirect

Environmental Modelling & Software

journal homepage: www.elsevier.com/locate/envsoft

Effectiveness of collaborative map-based decision support tools: Results of an experiment☆

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ARTICLE INFO

Article history:
Received 20 September 2011
Received in revised form
1 February 2012
Accepted 28 February 2012
Available online 27 March 2012

Keywords:
Collaborative decision support
MCA
GIS
Tool effectiveness
Support information
Land use planning

ABSTRACT

This article reports on the results of an empirical analysis of the effectiveness of a set of collaborative spatial decision support tools developed to support a land use allocation problem in a peat-meadow polder in the Netherlands. The tools feature spatial multicriteria analysis as the means to make spatially explicit trade-offs between stakeholder objectives in three different ways: as colors on multiple printed maps, qualitatively on a single digital map and quantitatively on a single digital map. An interactive mapping device is used as the interface between spatial information and participants. A series of controlled experiments was conducted with 30 participants, who were asked to use the tools and perform specific individual and group allocation tasks. Data on the responses of the participants were collected through questionnaires, observer notes, video film and multicriteria scores. The analysis focused on three aspects of effectiveness: usefulness of the tools; clarity of tool information; and impact on decisions. The findings of this analysis are discussed within the context of the information offered by the tools, individual and collaborative work of participants and the spatial and numerical quality of the resulting land use plans. From the experiments, it is clear that the cognitive effort related to the volume and format of information is a critical issue in spatial decision support. This holds true for both the level of detail per item of information and the amount of information to be supplied to the participants. Although the quantitative tool provided the most information, the participants did not consider this tool the most useful and it did not produce the best results.

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1. Introduction

Developments in natural resource scarcity, climate change, urbanization and population growth lead to enormous challenges to policy makers at various spatial levels in the 21st century. A number of conditions are to be satisfied in order to address these challenges in appropriate ways. In particular, it is important that institutions are strong enough so that communities can cope adequately with the conflicts that arise because of the above developments and that are also inherent to policies to address them. Moreover, sufficient knowledge on these developments and their consequences is obviously needed. In the present article, the focus is on the knowledge theme, but it is important to link it to the institutional dimension because knowledge can improve competences of decision-making bodies.

Little explanation is needed that the developments indicated above are interrelated and that their effects on the positions of various parties take place via complex mechanisms, where various disciplines (ecology, economics, social sciences), time scales and spatial scales have to be linked. This calls for integrated modeling approaches (see for example Argent (2004) for a review). As Argent indicates, many models start as stand-alone initiatives of researchers in specific disciplines to address specific problems. However, the need for integration and wider applicability gradually leads to the development of models with a wider scope, both in terms of linkages with other models and domains of possible applications. These developments usually run parallel to efforts to make them more user-friendly so that model developers step back and users are more strongly involved (Liu et al., 2008). In order to make integrated models useful for real-world policy applications, specific elements are often added to provide bridges between the world of models and the world of policy arenas. Among these elements are multicriteria analysis (MCA) and geographical information systems (GIS).

Multicriteria analysis is a decision support tool that contributes to a systematic comparison of policy alternatives in an integrated

☆ Thematic Issue on the Future of Integrated Modeling Science and Technology.

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doi:10.1016/j.envsoft.2012.02.021

manner. An important element is that it explicitly addresses conflicting objectives from various policy domains, such as agricultural productivity, nature and landscape. GIS is important since many of the decision problems relate to land use and land cover. Integrated analysis of environmental problems calls for a systematic accounting of land use change addressing a complete set of land use types. In an equal manner, from a policy domain land use is often a key policy instrument. GIS is also very helpful in terms of visualization of policy problems and their solutions (Argent, 2004).

The focus of this article is not on the development of integrated models as such, but on the fruitful ways of linking such models with real-world decision-making. This is done by using the tools of GIS and MCA in such a way that integrated models lead to knowledge that is presented in an integrated and intelligible way to policy makers working in a group context, implying some form of collaborative decision-making. More specifically, the article provides an in-depth test of the quality of various tools to convey the integrated knowledge generated by means of models to their users.

A common feature of collaborative map-based decision support tools is the use of maps as an interface between tools and end-users. These maps can be presented on paper as printed maps or on a screen, but are also increasingly being implemented in interactive mapping devices, such as the 'DiamondTouch Table' (http://www.circletwelve.com), the multi-touch 'Microsoft Surface' (http://www.microsoft.com/surface/), the SMART Board (http://www.smarttech.com), or interactive mobile devices such as GPS-enabled smart phones. Geertman and Stillwell (2009) distinguish between Planning Support Systems (PSS) and Spatial Decision Support Systems (SDSS). They state that PSS generally pay particular attention to long-term problems and strategic issues while SDSS are generally designed to support shorter-term policy making by individuals. They also state that SDSS deal primarily with operational decision-making while PSS focus more on strategic planning activities. Both PSS and SDSS combine tools from participatory geographic information systems (PGIS) with decision support tools, such as MCA and visualization tools (Eastman et al., 1998; Janssen and Herwijnen, 1998; Feick and Hall, 2002; Dragicevic and Balram, 2006; Malczewski, 2006). As explained above, MCA and GIS are promising tools to improve the effective use of output of integrated models since they potentially lead to integrated knowledge that is intelligible to the users.

Although many systems have been developed to date, the number of successful applications of both PSS and SDSS is still limited (Uran and Janssen, 2003; Geertman and Stillwell, 2009). In this light, it is surprising that the question of the effectiveness of these systems is seldom asked. Following the instrument approach of Goodhue (1995), two types of effectiveness can be distinguished:

1) How well the instruments enable users to carry out the intended tasks;
2) How well the instruments fit the capabilities and demands of intended users.

If the technology used in a system is relatively new, the Technology Acceptation Model (TAM) becomes relevant. Developed in the eighties by Davis (1989), the TAM model was originally designed to measure the factors that explain the acceptance and usage of classic information technology to the office environment, but has been since adapted to acceptance of wireless services and other types of information and communication technologies (ICT). This model predicts actual use of a device by its perceived usefulness and perceived ease of use. In this article, the focus is on the effectiveness of tools rather than to evaluate the technology. This article thus analyzes the effectiveness of a set of tools developed to support a land use allocation problem in a peat-meadow polder in the Netherlands. The tools were developed to support a series of planning workshops with governmental and non-governmental stakeholders. Results from these workshops, combined with observations and surveys, have shown that the tools are capable of supporting the planning tasks for which they were developed (see Arciniegas et al., 2011). Because the workshops were part of a political process, it was not possible to compare the situations with and without the tool or to compare different types of tools to support the same task. To facilitate drawing these comparisons and to test how well the instrument fits the capabilities and demands of intended users, a series of experiments was conducted.

In these experiments, three types of support tools were compared in ten sessions with a total of 30 participants, who were given specific individual and group tasks. The participants were observed while performing these tasks and answered survey questions after each task. This article describes the design and results of these experiments. The experiments focused on three specific aspects of tool effectiveness: 1) usefulness of the tool; 2) clarity of tool information; and 3) impact on decisions. Usefulness refers to the degree of success of a system in supporting a specific task or achieving a specific goal. Clarity refers to the ability of users to understand the information processed and presented by a tool. The impact of a tool on a decision process refers to the effects of the usage of a support tool on the decisions made by the intended users. This article is setup as follows: Section 2 includes a survey of literature dealing with the effectiveness of SDSS. The setup of the experiment is presented in Section 3. Results of the experiment are included in Sections 4 and 5. Conclusions are offered in Section 6.

2. Assessing effectiveness of collaborative SDSS

This section reviews recent empirical studies that focused on measuring the effectiveness (or several aspects of effectiveness) of collaborative SDSS, based on integrated models. Although there is currently a wide range of applications of this type of tools, not all of them are successful (Vonk et al., 2005; Geertman and Stillwell, 2009). There are a number of studies that reported on the reasons behind the success or failure of these tools. These studies focused on SDSS for coastal zone and water management (Uran and Janssen, 2003; Van Kouwen et al., 2008); PSS for urban and regional planning (Pettit and Wyatt, 2009); and environmental decision support systems for environmental management (Poch et al., 2004). Results of these comparative studies recommended a closer link between tool developers and end-users during the development stages of the systems to ensure that the tool supports the intended decision problem. In addition, a number of studies have pointed out the importance of testing whether this type of SDSS are effective and used (e.g. Jankowski and Nyerges, 2001; Uran and Janssen, 2003; Goosen et al., 2007; Vonk et al., 2007).

Previous studies have discussed various aspects of the effectiveness of collaborative SDSS (e.g. Sheppard and Meitner, 2005; Nyerges et al., 2006; Salter et al., 2009; Schroth et al., 2011). Effectiveness has been associated with the usability of a system in the context of human–computer interaction (Haklay and Tobón, 2003; Sidlar and Rinner, 2007; Meng and Malczewski, 2009). Moreover, effectiveness can relate to different aspects of a system or the tools included in it. From a broad perspective, three tool characteristics that are related to effectiveness of a tool can be identified: 1) the intended users of the tool; 2) the information included, processed and presented in the tool; and 3) the decision process to be supported by the tool and its outcomes. On the basis of these characteristics, this study distinguishes three individual aspects of effectiveness that can be evaluated separately, connecting each aspect to a specific question:

1) Usefulness: does the tool enable users to successfully perform the intended tasks?
2) Clarity: to what extent is the information presented in the tool understood by users?
3) Impact: how much do the tool and its outcomes influence the decisions made by users?

2.1. Usefulness of tools

Usefulness refers to the degree of success of a system in supporting a specific task or achieving a specific goal. Studies on usefulness generally involve sessions where participants are asked to interact with a tool to support the execution of tasks related to a certain spatial decision situation. Most of these studies deal with the degree of perceived usefulness, that is, the degree of satisfaction of users. Data on perceived usefulness can be collected by means of questionnaires or surveys (Dias, 2007; Voinov and Bousquet, 2010). For example, Hofstra et al. (2008) developed and tested an approach for collaborative decision-making that comprised mainly the usage of a multi-user touch-sensitive screen for disaster management in the Netherlands. This instrument allows simultaneous touches by multiple users and the use of maps, on which users carry out basic GIS tasks during the response phase after the occurrence of a disaster. This approach was tested in an experiment with 35 participants in order to assess its perceived usefulness and the perceived ease of use of the tool. Data were collected by means of a post-experiment questionnaire. The participants were asked to carry out a prepared assignment on a map of the city of Amsterdam. Results of the experiment show the high potential of the tool for disaster management and its suitability for a broad group of users. As a second example, Haklay and Zafiri (2008) used screenshots of results achieved using a tool to measure successfulness. Further examples of empirical studies that aimed to measure the usefulness of tools can be found in Andrienko et al. (2003), Janssen and Uran (2003), Goosen et al. (2007), Sidlar and Rinner (2007) and Inman et al. (2011).

In an empirical study, Sidlar and Rinner (2009) assessed the successfulness of a map-based online geo-collaboration tool. The authors measured how well the tool facilitated an online geographically referenced discussion. The tool consisted of a web-based mapping application, with which users can provide feedback on land use plans by pointing to places on a map in combination with an on online discussion forum. An experiment was conducted with 11 participants. The degree of success of the tool was calculated as a 'utility ratio', which is the ratio of the actual use of the tool (counts obtained from web server log files) over the potential use (number of geographically referenced contributions made to the discussion forum). The success of a tool can also be measured by tracking user–computer interaction. For example, Meng and Malczewski (2009) demonstrated how to quantitatively evaluate the usability of a web-based public participatory GIS called 'ArgooMap'. ArgooMap comprised an MCA module to calculate the scores of all alternatives and a 'Group Decision Component', with which alternatives are ranked based on the majority's preferences. The experimental part of this study examined the relationships between user characteristics and system usability. In this study, usability was regarded as the easiness of using these systems for a wide range of users and was defined in terms of the users' performance, which involved effectiveness, learnability, efficiency and user satisfaction after interacting with the tool. Tourists and local residents were invited to use the tool and asked to identify preferences regarding the location of new parking places. Post-questionnaires and a tool called 'UsaProxy' were used to collect data in order to test the usefulness of the tool. UsaProxy tracks and records all interactions between the user and the tool remotely. Data were analyzed with descriptive statistics and by correlating user characteristics and successfulness metrics. Boroushaki and Malczewski (2010) used a quantitative measuring of the degree of consensus achieved by participants as an indicator of the effectiveness of an online collaborative tool.

The usefulness of a tool can also relate to both its functionality and usability. For example, a recent study by Vonk and Ligtenberg (2010) included the evaluation of two PSS for sketch planning. The authors defined functionality as the match between the planning tasks and the system. Usability was defined as the match between user needs and the PSS. The goal of this evaluation was to compare traditional PSS development methods that involved only PSS developers with an approach that involved a close collaboration between direct users and PSS developers. Both PSS integrate drawing and sketching with basic GIS and a tabletop called the 'Maptable'. Their evaluation comprised aspects, such as functionality, usability and the relative advantage between traditional methods and a socio-technical development approach. Eight expert observers filled out an evaluation form with questions and criteria. Both the planning and evaluation processes were filmed. Data were obtained from observations and the comments of the experts. Results from the evaluation suggested that cooperation between users and programmers of PSS during development stages contributes to an increased functionality and usability. Vonk and Ligtenberg (2010) recommended further testing these systems, particularly those featuring tabletop technology, for usability for various types of planning tasks, processes and user groups. Pettit et al. (2011) assessed strengths and weaknesses of landscape visualization tools, namely virtual globes, for communicating alternative landscape scenarios in collaborative environmental planning. A number of alternative landscape scenarios were communicated to two types of audiences: current users (12 environmental managers and planners) and future users (89 undergraduate spatial information students). Interviews and web-based surveys were used on each audience respectively to assess the tools and to derive suggestions for tool improvement.

2.2. Clarity of information

Clarity refers to the ability of users to understand the information provided by a tool. It does not necessarily relate to the design of the tool or the models used to generate the information. As an example, Janssen and Uran (2003) used interviews to analyze how well stakeholders understood the results of an SDSS for water management in the Netherlands. The stakeholders were interviewed to determine their preference for maps, text, or graphs as a means of communication and to analyze the relationships between level of detail, perceived confidence, difficulty, correctness and the perceived value of maps for decision-making. An interesting result was that although participants indicated that they preferred fairly detailed maps, this did not always coincide with their ability to use these maps. Goosen et al. (2007) analyzed the use of a negotiation support tool for regional water management in the Netherlands. Information presented included valuation tables, value maps, conflict maps and the performance score for each objective calculated with MCA. The participants worked in three parallel groups and were asked to represent different stakeholder interests of a wetland area in the Netherlands. After each workshop, the participants filled out a questionnaire on their understanding of the tool and the information involved. Results from these surveys showed that the participants found the value maps easier to understand than the aggregated value scores.

Kennedy and Bishop (2008) reported on six experiments that aimed to measure participants' land use knowledge and choice

behavior when using a virtual decision-making environment. A virtual environment integrates 3D visualization and results from MCA to assess relative impacts of different land use scenarios. Twenty-nine people participated in the experiment and were asked to play the role of a farmer and manage three parcels of land for the next twenty-five years. They were then asked to make land use choices under six different experimental modes. The modes differed in type of information presented (2D and 3D) and decision conditions (in isolation and seeing neighbor's parcels). After the sessions, the users were asked to fill out questionnaires in order to rate their experience and knowledge. Results were of limited statistical significance as to what extent knowledge of choices of participants using the virtual environment influences land use choice behavior. Consequently, further testing with refined controlled conditions was recommended. Several other empirical studies stressed the importance of clarity of a decision support tool (Andrienko et al., 2003; Bacic et al., 2006; Sidlar and Rinner, 2007; Pettit et al., 2011). Particularly, Andrienko et al. (2003) discussed the importance of promoting a clear understanding of how the results are obtained and the existence of different styles of decision-making that require distinct support tools. Similarly, Bacic et al. (2006) stressed the relevance of the role of visualization to increase the stakeholder's understanding of a decision problem.

2.3. Impact on decision process

The impact of a tool on a decision process refers to the effect the use of a support tool has on the decisions made. The effect can be positive, negative, or absent. The use of a tool can also influence decisions to varying degrees. Impact can be reflected on the results achieved by users, or can be related to the extent, frequency, and so on, of information use that lead to decisions. There are several approaches in the literature that utilized different indicators to measure the impact of SDSS. Nyerges et al. (2006) tested a collaborative SDSS for ground water and surface water systems called 'WaterGroup'. The goal of the experiment was to evaluate the impact of the SDSS on the collaborative decision process. The authors' definition of impact referred to the ability of the tool to achieve 'meaningful stakeholder participation' after providing stakeholders with information about water resources within a collaborative decision situation. The decision process was characterized in terms of human–computer–human interaction and the experiment goals were framed into a number of research questions. The tool was tested with two groups of stakeholders in a controlled experiment consisting of two workshop sessions, each group with a different software-facilitator configuration. Participants were asked to use the tool to create and select a water resource management plan for the region. Participant interaction and software use were registered with video cameras and computer activity logging. All participants were asked to fill out questionnaires before and after the sessions to record their perception on the software and the decision process. Results of this experiment showed that different technology configurations encourage a different balance between analysis and deliberation activities of spatial decision-making processes.

In an empirical study, Dias (2007) utilized an assessment methodology framework to measure and quantify the impacts and added value of contextual mobile spatial information services on 416 visitors of natural areas. This methodology involved comparing different information dimensions (e.g., having the information or not) and different information mediums. Pre and post-questionnaires were used to collect data on the spatial behavior, perceptions and opinions. Impact was measured with contingent valuation of information using preferences stated by the participants. This study found that the perception of added value increases when tool information becomes more detailed.

A study by Salter et al. (2009) aimed to evaluate the impact of a participatory planning workshop. The authors categorized effectiveness into three aspects: 1) effectiveness of the workshop sessions; 2) effectiveness of the interactive spatial decision support tools for landscape planning and its components; and 3) participants' feedback on the resulting plans. Planning workshops were held with experts involved in the planning process. In the workshops, the participants had the opportunity to explore, discuss and numerically assess different landscape scenarios on several indicators using an interactive planning tool. This evaluation was conducted using data collected through three methods: 1) pre and post-session questionnaire on the helpfulness of the workshop components and the tools; 2) video analysis; and 3) researcher observations. Participants' input suggested that the tool and its components helped them increase their understanding of the resulting plans. Salter et al. (2009) concluded that this type of mixed technology for tool and process evaluation constitutes a potentially useful means for measuring participant interactions in this type of workshops. A similar study by Sheppard and Meitner (2005) reported on the evaluation of an MCA-based SDSS by stakeholders of a sustainable forest management planning process. All participants were asked to provide feedback on the planning process and its individual components, such as focus group, criteria weighing, scenario evaluation and visualization. Martinez-Santos et al. (2010) discussed strengths and weaknesses of two tools for participatory water management with respect to each stage of participatory integrated assessment. The tools, namely Bayesian belief networks and hydrological modeling, were tested on a stakeholder group of a ground water management process in the Mancha Occidental aquifer, Spain. An important conclusion is that a combination of the tools is more desirable for process support that each tool alone, since the former tool was considered more valuable for prompting stakeholder dialog and the latter for communication of information. Other studies that examined the impact of spatial information and tools on a decision process agree that the degree of interactivity of a tool is an important aspect that can influence decision-making (e.g. Andrienko and Andrienko, 2003; Bacic et al., 2006; Goosen et al., 2007).

This section presented a number of empirical studies that focused on measuring the effectiveness of collaborative SDSS. These studies used a range of data acquisition methods such as surveys, questionnaires, observation, video analysis, screenshot analysis and user activity tracking techniques. A common aspect of the majority of these studies is that the assessments were largely based on user opinion and perception. Only a couple of studies actually included quantitative performance measures. Hence, this review suggests that there is a shortage in the literature of studies on the quantitative measurement of effectiveness of SDSS. The following section describes the empirical part of the present study, which involves the analysis and comparison of qualitative and quantitative measures on the basis of the three aspects of effectiveness.

3. Design of the experiment

3.1. Supporting collaborative planning for a polder in the Netherlands

The Bodegraven polder (Fig. 1) is a low-lying peat-meadow area (an average of 2 m below sea level) of 4072 ha where water tables are controlled to enable different types of land use. Bodegraven is currently facing several problems: ground subsidence; gradual disappearance of the peat-meadow landscape; inefficient water

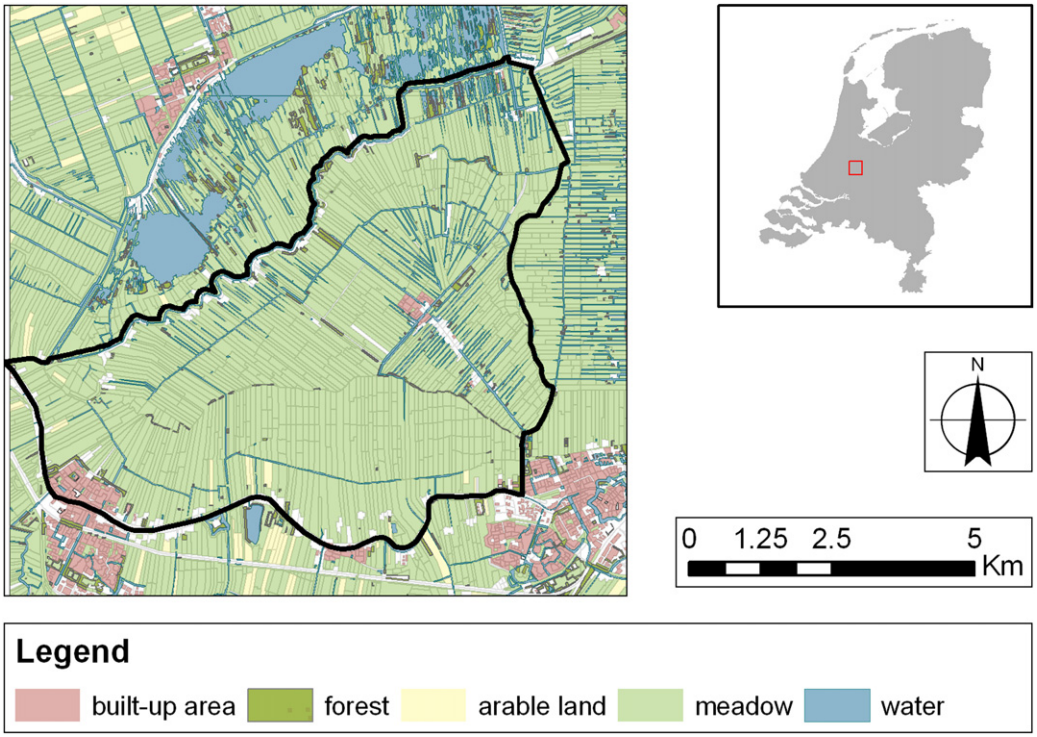

Fig. 1. The Bodegraven Polder, The Netherlands. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

management; poor water quality; and the declining economics of dairy farming. Consequently, the provincial authorities started a process to review and adjust both water management and land uses in the area.

The support of the planning process was structured in three interconnected workshops. The starting point was the design workshop, where the playing field was defined. This involved: 1) the identification of the range of relevant alternatives; and 2) an inventory of the objectives of the various stakeholders and the evaluation criteria linked to these objectives. Four objectives were identified: 1) profitability of agriculture; 2) minimization of land subsidence; 3) maximization of landscape quality; and 4) maximization of natural value. Three land use types were identified: intensive agriculture, extensive agriculture and nature. Participants of this workshop were public stakeholders. The second workshop focused on the assessment of the criterion and objective values of the different types of possible land uses in the area. These values were presented as value maps. Participants of this workshop were experts with local knowledge. The value maps produced in the first two workshops were the input for the third workshop, the negotiation workshop. This last workshop supported the process of collectively changing a reference plan into a new negotiated plan. Participants of this workshop were members of the project group commissioned to develop a plan for the region. Prior to the workshop, a series of experiments using MSc students as participants was conducted to test the negotiation tools. For these tests, real stakeholders were not used for two reasons: first, with real stakeholders is not feasible to test different types of support; and second, with real stakeholders, it is difficult to distinguish between the influence of specific knowledge of these stakeholders and the influence of the tool.

3.2. Decision support tools

Three decision support tools were tested. The tools provided users with feedback information in three different ways: 1) as colors on multiple printed maps; 2) quantitatively on one digital map using numbers; and 3) qualitatively on one digital map using letters. The tools were developed with CommunityViz Scenario 360 (http://www.communityviz.com/). Technical details can be found in Arciniegas et al. (2011). In all three tools, spatial MCA was used to show trade-offs between objectives across the area. Weighted summation was used to aggregate standardized criterion values into performance scores for objectives and an overall performance score. To calculate the objective values, criterion values are first standardized between 0 and 10 and then multiplied by their weights. Next, the objective values are obtained as the sum of these products. The total value is calculated as the weighted sum of the objective scores (Janssen, 1992). The weights represent the relative importance of the criteria or objectives. Criterion weights were set and specified using expert judgment. Equal objective weights were used to calculate the total value in Fig. 2. These weights can be adjusted during the workshop. In this application, the objective and total values are first calculated for each parcel. This results in value

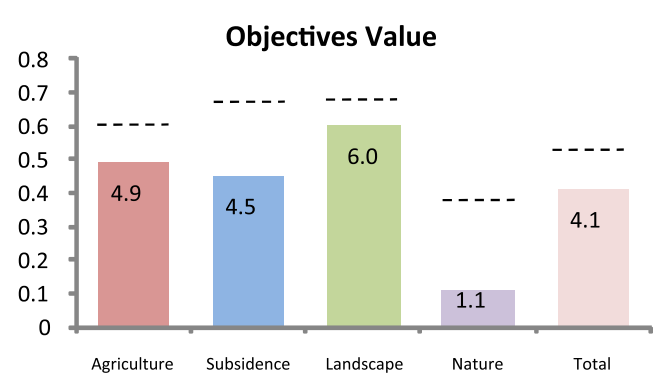

Fig. 2. Objective values and Total MCA value.

maps for each objective and a total value map. The values for the area as a whole (as presented in Fig. 2) are calculated as the area weighted average of the values of all parcels. Stakeholders used the tools to identify trade-offs, agree on an exchange of land use and change land use patterns on the Touch table. To change the land use of one of more parcels, stakeholders use their fingers to select a land use class and then allocate it to selected parcel(s) on the Touch table. Fig. 2 shows the values of the four objectives, as well as the total value for the entire study area. The dashed lines on each bar indicate the single objective maximum for each objective. If the participants changed the land use of one or more parcels, the values were automatically reassessed in real time. This information was available in combination with the different support tools on a separate screen.

3.2.1. Tool 1: multiple-map colors

This tool consists of three choropleth classified maps printed in A3 format (Fig. 3). Each map shows a classification of standardized objective values into ten equal classes, which are depicted in a red-to-green color ramp and rendered on a base polygon map (scale 1:50,000) representing land parcels in the polder. The first of the three maps shows the potential value of the nature objective if the land use for the whole area would be nature. This map shows the best parcels for nature (green), the worst parcels for nature (red) and parcels in between. The second and third maps do the same for extensive and intensive agriculture. The three maps were available to all participants of a session. Participants can use these maps to support the identification of parcels that may profit from a land use exchange. Trade-offs between objectives can be identified, for example by looking at two or more maps at the same time.

3.2.2. Tool 2: single-map quantitative

This tool presents the objective values that are linked to the three types of land use in one digital base land use map (1:50,000) on the 'Touch table'. The base polygon map shows a land use classification into three categories, which represent the current land use of a parcel. The values for the three objectives are shown on each parcel as a three-number label as follows: the first number represents the value of the parcel if it would be used for nature; the second for extensive agriculture; and the third intensive agriculture (see Fig. 4). For example, the parcel in the top right of Fig. 4 is currently in use for intensive agriculture. The three numbers in the parcel (7-7-4) indicate that the value is low for intensive agriculture (4) and would improve if it was changed to nature (7) or extensive agriculture (7). It is assumed that stakeholders try to obtain parcels that have a high value in exchange for a parcel with a low value. On the map, this means that participants will look for high/low and low/high combinations to identify parcels that are candidates for a favorable exchange. While most parcels in Fig. 4 have land use intensive agriculture, not all of these parcels have a high value for intensive agriculture. If this low value goes together with a high value for another land use, this parcel becomes a good candidate for exchange. To support negotiation, the tool highlights trade-offs between objectives by coloring the boundaries of those parcels that are potentially favorable for an exchange of two land use types. In order to highlight trade-off information, users utilize an interface which appears on the Touch table on request. The interface has a number of interactive sliders, with which users can specify high or low thresholds for scores of each objective. Thresholds are set by touching and dragging one slider to a desired position. For example, the interface can be used to highlight parcels holding nature values higher than 8 and values for intensive agriculture lower than 1. Fig. 4 shows that, based on this setting, a parcel with values 8-7-1 gets its boundaries colored in pink. This tool supplies the most information of all three.

3.2.3. Tool 3: single-map qualitative

This tool provides qualitative feedback on a digital land use map on the 'Touch table'. With this tool, participants can select and identify the best or worst parcels for each land use on the basis of the objective values. The tool features an interface, similar to that of the single-map quantitative tool, which appears on the Touch table on request and has interactive sliders for each objective. By touching a slider and dragging it to a new position, users specify a percentage of the study area (or the equivalent in hectares) to be analyzed and ranked on the basis of the objective values. For example, depending on the assignment, participants can use the tool to highlight the best 20% of all parcels with the highest score for land use 'nature' and the 20% with the lowest scores for land use 'intensive agriculture'. In each parcel, a sequence of one, two or three characters is displayed representing the objective values associated with the three land uses. This tool is based on the assumption that participants will look for high/low and low/high combinations to identify parcels that are candidates for favorable exchange of land use between participants. With the tool, participants specify the size of the area in hectares to be included in the negotiation. For each land use, the tool selects the relatively best and worst parcels adding up to the specified area size. A fixed sequence of three characters, with each character representing a land use type, is used to identify selected parcels as follows: good (high value) parcels are labeled with blue characters and bad (low value) with red (Fig. 5). The negotiation map shows the best (blue) and worst (red) parcels for each of the three land uses: Nature (N), Extensive Agriculture (E) and Intensive Agriculture (I). For example, a sequence of characters 'N E I', colored respectively blue, blue and red, indicates that a parcel falls within the best area for

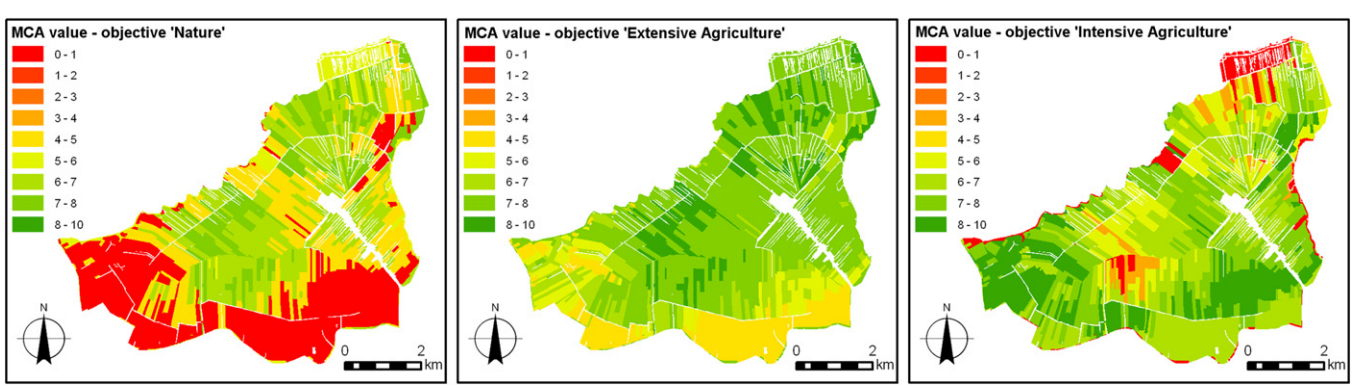

Fig. 3. Feedback as colors on multiple objective-specific printed maps. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

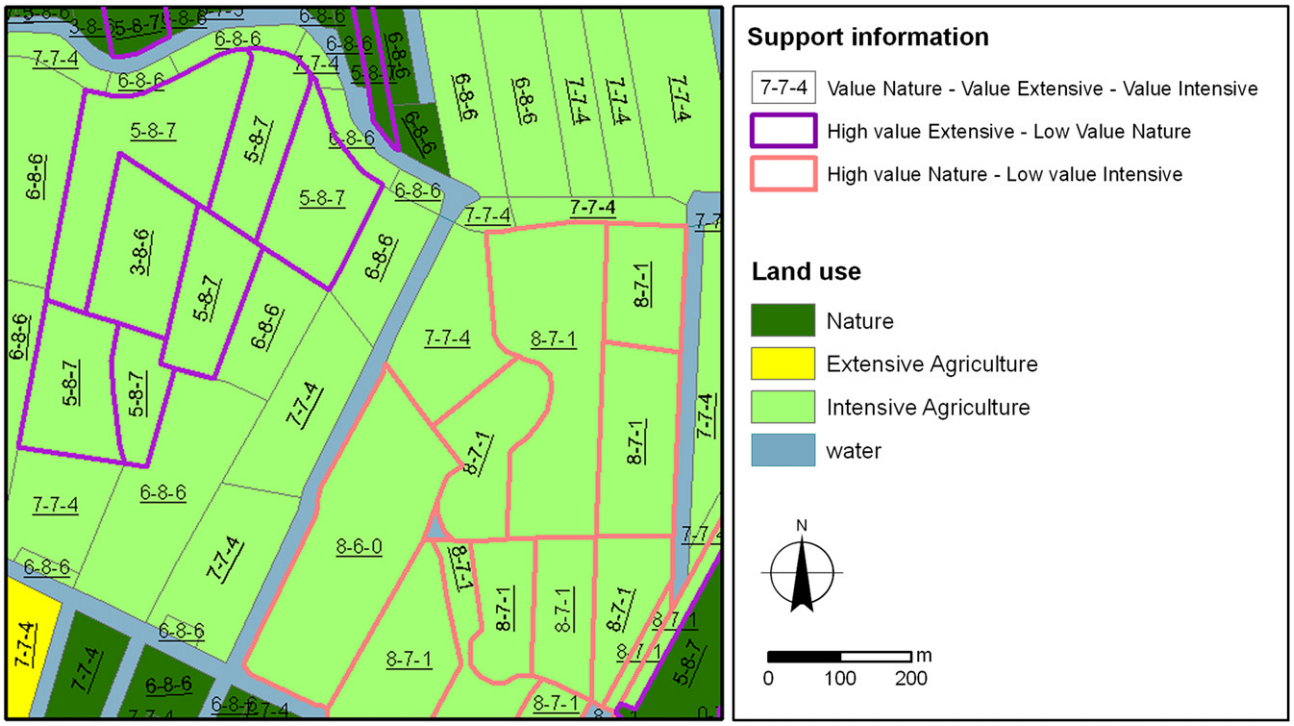

Fig. 4. Quantitative feedback on a single digital map. Trade-offs are highlighted with color-coded boundaries. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Nature and Extensive agriculture and within the worst for Intensive agriculture. If these parcels are currently in use for intensive agriculture, they make good candidates for exchange. Parcels that are not within the specified limits are left blank.

3.3. Hypotheses of experiment

Four hypotheses are tested to evaluate the effectiveness of the three support tools:

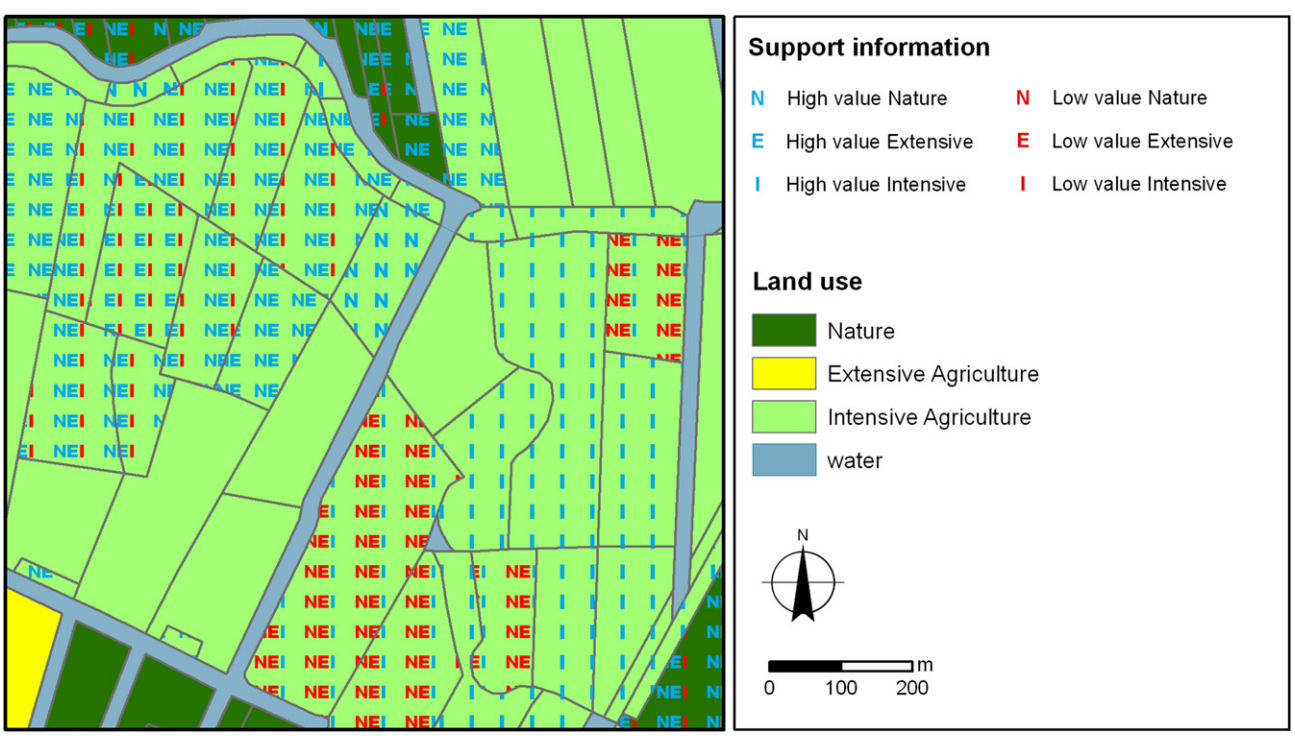

Fig. 5. Qualitative feedback on digital maps. Trade-offs are highlighted with color-coded labels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Hypothesis 1. The perceptions of tool effectiveness run parallel to the performance outcomes on a group level.

Hypothesis 2. The amount of information supplied in the tools affects the participants' understanding of the tools.

Hypothesis 3. The intensity of tool use affects the performance outcomes on a group level.

Hypothesis 4. The individual performance in using the tools influences the outcomes on a group level.

3.4. Setup of the experiment

3.4.1. Participants

Thirty people participated in the experiment. All participants were international students from the MSc program in environment and resource management of the VU University Amsterdam. The majority of participants had an academic background in economics, environmental management, biology, or international relations. Participants were asked to carry out an assignment three times. Each time, they used a different support tool in an order defined randomly. For each tool there was both an individual and a group assignment. At the end of each assignment, they were asked to rate their experience with the tool and to answer a couple of questions related to their understanding of the tool and the information used.

3.4.2. Hardware

The layout of the experimental session is illustrated in Fig. 6. Three participants were seated at the Touch table (display area of 86 cm × 65 cm), which was the interface to the tools. The graph of Fig. 2 showing the objective and total values for the whole area was displayed on a separate screen. An observer stood beside the Touch table. A digital video camera recorded all sessions and stood to the left of the screen capturing all interaction among participants and between them and the Touch table. This experimental setup builds upon the format of earlier studies, such as Nyerges et al. (2006) or Salter et al. (2009), in which cameras, observations, questionnaires, or computer activity logging were used to record interactions between participants and systems in a workshop setting.

3.4.3. The assignment

The experiment involved ten 2.5-h sessions. Each session included a group of three participants. These participants were asked to complete both a group and an individual assignment. The assignments were repeated for all three tools. In the group, each participant was asked to play a stakeholder role from the three possible roles (nature organizations, agricultural nature organizations, farmers' organizations) and increase the quality of a specific objective (Nature, Landscape, Agriculture). At the same time, each group was asked to increase the total quality of the reference plan as they tried to achieve long-term provincial policy goals in hectares for each land use. The group assignment was to:

- Allocate 400 ha of land use *Nature*
- Allocate 400 ha of land use *Extensive Agriculture* and 2755 ha of *Intensive Agriculture*
- Maximize the values of the three objectives and the *Total* value
- Try to connect parcels with land use *Nature*

The individual assignment was a simplification of the group assignment. Each participant was asked to use a tool separately to modify a land use situation within a subset of the study area. Specifically, each participant was asked to play the role of the stakeholder 'Nature organizations' and within 5 min relocate 100 ha of land use *Nature* reaching the following goals:

- Relocate 100 ha of *Nature*
- Maximize the *Nature* value and the *Total* value
- Try to connect parcels with land use *Nature*

The timeline of one experimental session is illustrated in Fig. 7. The session began with an introduction to the study area and a short explanation of the tools followed by some hands-on practice with the tool. Next, both the individual and group assignments were explained, followed by a pre-session questionnaire. After introducing the first tool, each participant was asked to carry out the individual assignment using the first tool, one participant at a time. As soon as the third participant completed the assignment, all three were asked to collectively carry out the group assignment using the first tool. After completion, all participants were asked to individually fill out a questionnaire about their perception on the effectiveness of the tool and the entire system and to test their knowledge of the tool with which they just worked. These steps were repeated for the second and third tool. Once the work with the third tool was completed, a post-session questionnaire was administered to the participants.

The questionnaire records the effectiveness as perceived by the participants. The actual performance of the participants was monitored by means of an observer, video film, MCA scores and spatial patterns of plans. The observer kept track of the activities of the participants during the individual assignment and the interaction between participants during the group assignment. For example, the observer was to check whether or not a participant chose the right map or correctly used a support tool to complete an individual task, or kept track of how many times a group referred verbally to the information included in a tool in order to discuss one idea. Video film was used to acquire data on conflict management, time spent using support tools and time spent discussing. MCA scores and spatial patterns were used as an indicator of the quality of the plans generated by the participants.

This assignment can be considered a form of scenario generation and evaluation. The products of both individual and group assignments are a set of future land use scenarios for the study area. Individual participants use the tools to generate scenarios for a section of the polder, while the groups generate collectively scenarios for the entire polder. All scenarios are evaluated using MCA on the basis of stakeholder objectives, as described in Section 3.2. Similar studies have included scenario evaluation assignments to test SDSS (e.g., Pettit, 2005; Kennedy and Bishop, 2008; Salter et al., 2009).

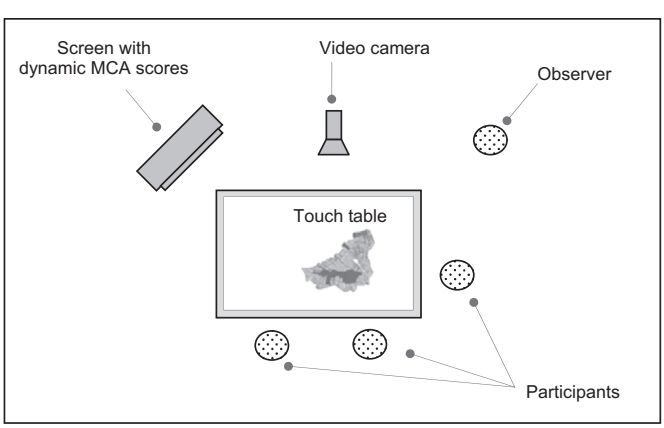

Fig. 6. Arrangement of participants and hardware for the sessions.

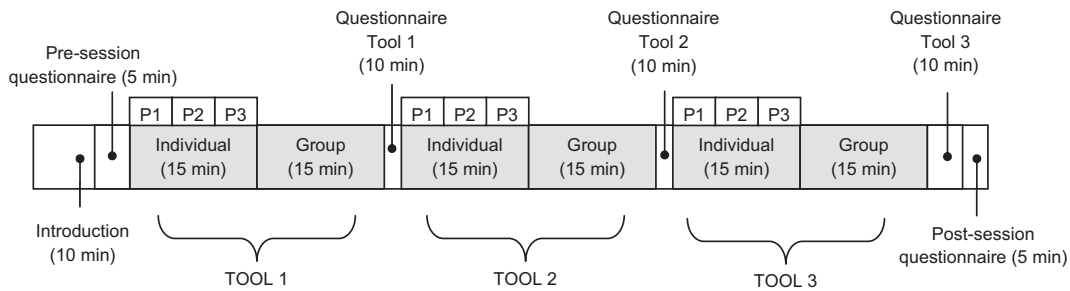

Fig. 7. Timeline representing the experiment protocol of one session.

3.4.4. Limitations

The present empirical approach has three main limitations. First, using students as test subjects instead of real stakeholders can be an issue mainly for two reasons, namely a lack of local knowledge and a younger age, which implies a presumably higher acceptance of the technology being tested. However, the fact that students have neither local knowledge nor stakes concerning the study area can contribute to a more transparent evaluation of tool effectiveness, since it excludes the effect of local knowledge on the decisions. This effect might in fact be measured by comparing results obtained with students against those obtained with real stakeholders (e.g., in Pettit et al., 2011) or by comparing results obtained with and without the tool. In addition, while students can be more willing to embrace new technology, survey results from other studies that tested similar technologies with stakeholders relatively older and/or with less education than MSc students revealed no significant difference in the acceptance of the technology (e.g., Alexander et al., 2012; Arciniegas et al., 2011). Second, having the same group of participants work with all three tools and the sequence of tool exposure could have had an effect on the outcomes that could be associated with carryover and learning effects (e.g., they might have become more skilled and efficient with every assignment). It would have been ideal to include all possible sequences of tool exposure in the analysis, allowing sufficiently long periods between each tool exposure, so that participants can disassociate themselves with the assignments and tools they just worked with. However, this proved to be impossible to realize due to limitations of time and student availability. Third, the relatively small sample size (30) might have influenced the precision of results. A larger sample size would have provided more statistical significance. Although online access to tools could have allowed for larger audiences (e.g., Meng and Malczewski, 2009), the present approach considers personal interaction an essential aspect of the tools tested. The hardware setup of the land use allocation tools of this approach required participants to interact with each other to allow give-and-take around the Touch table.

4. Results of experiment

Of the thirty subjects who took part in the experiments, 23% have a background in economics, 17% in environmental management, 10% in international relations and 10% in biology. Furthermore, 24% have some or considerable experience with digital maps, 16% with GIS and 48% with MCA.

4.1. Individual measurements

Data on individual perceptions regarding the tools were acquired from the participants' responses to the questionnaires. These responses are organized on the basis of the three aspects of effectiveness (F1. Usefulness; F2. Clarity; and F3. Impact). Table 1 summarizes these responses, indicating scales of measurement for every indicator (I) and detailed indicators (D). Participants rated the tools by specifying one level of satisfaction for a specific aspect using a 7-level Likert scale. This rating scale has three negative levels of satisfaction (−3, −2, −1), one middle neutral level (0) and three positive levels (+1, +2, +3). Such a scale has been used in previous empirical studies on tool effectiveness (e.g., Dias, 2007; Inman et al., 2011).

Data on individual performance is organized on the basis of the three aspects of effectiveness (F1. Usefulness; F2. Clarity; F3. Impact) (Table 2). The first aspect (F1) refers to the ability of a participant to execute a number of individual tasks necessary to complete a favorable land use exchange. Data were acquired by observer notes, which indicate whether or not a participant successfully completed a specific task. The second aspect (F2) shows whether or not a participant answered correctly a multiple-choice question about the information included in each tool. Concerning tool 1, participants were asked the following question: *what does it mean that one portion of the study area appears as red on one map and green on the other?* Concerning tool 2 and 3, participants were asked the same question twice: *by setting values with the interface, a number of parcels are highlighted on the map. On which basis are these parcels selected?* The third aspect (F3) comprises two measurements: Total MCA scores of plans and the number of parcels with a new land use.

Table 1
Average perceived effectiveness per tool. A figure in bold denotes maximum, in italics denotes minimum.

		Multiple-map colors	Single-map quantitative	Single-map qualitative
F1. Usefulness of tool (questionnaire)				
I1. Usefulness of tool	Likert: −3 to +3	2.267	*1.600*	**2.483**
D1. Usefulness to identify trade-offs	Likert: −3 to +3	**1.867**	*1.414*	1.767
D2. Usefulness to support swaps	Likert: −3 to +3	2.000	*1.345*	**2.067**
D3. Usefulness to address conflict	Likert: −3 to +3	1.900	*1.036*	**1.931**
F2. Clarity of tool (questionnaire)				
I1. Clarity of tool information	Likert: −3 to +3	2.067	*1.700*	**2.267**
F3. Impact on decision process (questionnaire)				
I1. Valuation of tool	Likert: −3 to +3	1.222	*1.037*	**2.464**
I2. Influence of tool on decisions	Likert: −3 to +3	1.593	*1.519*	**2.407**
I3. Performance of plan	Single choice: %	*3.6*	17.9	**78.6**
Preferred support tool	Single choice: %	*10.7*	21.4	**67.9**
Overall perception	Likert: −3 to +3	1.700	*1.378*	**2.119**

Table 2
Average observed individual performance per tool. A figure in bold denotes maximum, in italics denotes minimum.

		Multiple-map colors	Single-map quantitative	Single-map qualitative
F1. Usefulness of tool (observer notes)				
D1. Identify trade-offs	yes = 1/no = 0	33.3%	**70.0%**	*25.9%*
Identify own valuable parcels	yes = 1/no = 0	*83.3%*	**86.7%**	**86.7%**
Identify others' valuable parcels	yes = 1/no = 0	*40.0%*	60.0%	**86.7%**
D2. Performance swaps				
Negotiate swaps	yes = 1/no = 0	*40.0%*	*40.0%*	**96.7%**
Carry out swap	yes = 1/no = 0	*40.0%*	*40.0%*	**60.0%**
D3. Performance conflict	yes = 1/no = 0	**80%**	*40%*	*40%*
F2. Clarity of tool information (questionnaire)				
C1. Test of subject's understanding	yes = 1/no = 0	*36.7%*	**80.0%**	70.0%
F3. Impact on decisions (MCA scores and spatial pattern)				
C1. MCA score normalized by min-max range[a]: 4.17–4.40	0 to 10	*1.568*	4.490	**5.325**
Increase in MCA score; reference = 4.17	0 to 10	*0.036*	0.103	**0.122**
C2. Parcels with a new land use	Count	**57.33**	*40.30*	44.20

[a] Normalization was performed using 4.17 as the baseline Total Score of the reference situation. The Score of 4.40 was obtained through linear optimization using the area goal of the individual assignment.

According to the results in Table 1, the participants found that the single-map qualitative tool was the most useful, the clearest and had the highest impact on their decisions. Conversely, the single-map quantitative tool received the worst valuations on these three aspects. The qualitative tool was considered to be the best for supporting all tasks, except 'identify trade-offs', which was thought to be best supported by the multiple-map colors tool. The overall perception was that the qualitative tool was the best and the quantitative the worst. According to observed performance (Table 2), the qualitative tool allowed for the best individual performance in the execution of all tasks, except 'identify trade-offs', which was best supported by the quantitative tool. The multiple-map color tool was not really understood by the participants, as only 36.7% gave a correct answer. Both the quantitative and qualitative tools were understood by 80% and 70% of the participants, respectively. MCA results show that the qualitative tool helped participants produce the plans with the highest MCA value, whereas the multiple-map color tool helped produce plans with the lowest value.

4.2. Group measurements

Data on group performance were collected from observer notes, video footage and the land use plans generated by the groups. Table 3 presents these measurements on the basis of the three aspects of effectiveness (F1. Usefulness, F2. Clarity and F3. Impact). Data on usefulness were obtained from the resulting land use plans as the Total MCA score of each plan. Data on clarity originated from the participants' answers to the test of knowledge. Data on impact are presented in five categories: 1) intensity of collective tool use;

Table 3
Observed average group performance per tool. A figure in bold denotes maximum, in italics denotes minimum.

		Multiple-map colors	Single-map quantitative	Single-map qualitative
F1. Usefulness of information (MCA scores)				
C1. MCA score normalized by range[a]4.11–4.52	0 to 10	6.10	*5.12*	7.32
F2. Clarity of tool information (questionnaire)				
C1. Test of subject's understanding	yes = 1/no = 0	*36.7%*	**80.0%**	70.0%
F3. Impact on decisions				
C2-a. Intensity of tool use (observer notes)				
Discussed ideas	Count	*12.30*	14.20	**16.78**
Use of tool information	Count	13.70	*6.60*	**15.60**
Change land use	Count	13.10	*8.90*	**13.20**
Using chart with MCA scores	Count	8.10	*7.90*	**8.30**
C2-b. Time using tool (video)				
Time spent discussing	min:sec	**4:25**	*2:14*	2:48
Time spent working on Touch table	min:sec	**10:50**	*8:25*	9:18
Time spent using chart MCA scores	min:sec	**1:13**	*0:46*	0:58
C2-c. Group strategy (observer notes)				
Group discusses strategy before starting	yes = 1/no = 0	*60%*	*60%*	**80%**
Group refers to information presented in tool	yes = 1/no = 0	**100%**	*90%*	*90%*
Group refers to MCA information on screen	yes = 1/no = 0	**80%**	67%	*56%*
Two participants negotiate at once	yes = 1/no = 0	**40%**	30%	*20%*
Three participants negotiate at once	yes = 1/no = 0	*60%*	**70%**	*60%*
C2-d. Performance conflict (observer and video)				
Conflict emerged	yes = 1/no = 0	**80%**	*40%*	*40%*
Information of tool considered	yes = 1/no = 0	**100%**	*75%*	**100%**
Extent of use of tool	0 to 3	**1.90**	*0.60*	1.00
C3. MCA scores and spatial pattern				
C3-a. MCA score normalized by range[a]4.11–4.52	0 to 10	6.10	*5.12*	**7.32**
Increase in Total MCA score; reference = 4.11	0 to 10	0.25	*0.21*	**0.30**
C3-b. Number of parcels with a new land use	Count	**303.1**	*202.0*	261.0
C3-c. Spatial pattern of land use map (clustering)	−1 to +1	**0.308**	*0.272*	0.289

[a] Normalization was performed using 4.11 as the baseline Total Score of the reference situation. The Score of 4.52 is the theoretical maximum value obtained through linear optimization of that score using the area goals of the group assignment.

2) time measurements; 3) group strategy; 4) conflict management; and 5) performance of plans generated.

Data on intensity of information use (C2-a) were acquired from observer notes and organized into four indicators. The observer counted how many times a specific task was carried out; for example, how many times the group consulted the tool to display trade-offs on the map, or how many times new land uses were reallocated to parcels. Time measurements (C2-b) were acquired from video footage and included two indicators for tool use and one for discussion. Data on group strategy (C2-c) were obtained from observer notes and organized into five indicators. For example, it was noted whether or not the group defined the strategy before starting the assignment or whether the decisions were made on a two-participant or a three-participant basis. Conflict management (C2-d) included three indicators, acquired from observer notes and video: whether or not conflict arose, whether the tool was used to address it and the extent of this usage. Plan performance (C3) comprises three indicators, which were obtained from the resulting land use plans: Total MCA scores of the plans (C3-a), the number of parcels with a modified land use (C3-b) and land use spatial patterns (C3-c). Table 3 shows that, in terms of intensity of use, the single-map qualitative tool was used the most intensively and also allowed for the most ideas discussed among participants. Time measurements show that the multiple-map color tool was used for the longest time and also prompted the longest discussions. As for group strategy, the multiple-map colors tool allowed for the highest scores for the most indicators. In terms of conflict management, the multiple-map color tool performed best. MCA results show that the groups generated the best plans with the qualitative tool and the worst with the quantitative tool.

5. Comparing perceived and observed effectiveness

5.1. Indicators of effectiveness for individual and group decision-making

This study analyzes the effectiveness of decision support tools on the basis of three dimensions: F1. Usefulness, F2. Clarity and F3. Impact. Each dimension is in turn measured with a number of indicators and validated through comparisons against observed measurements, or controls. Table 4 and Table 5 contain two frameworks for individual and group effectiveness, respectively, illustrating how each dimension is represented by indicators.

To test how well the three dimensions together contribute to measure overall perceived effectiveness, a reliability analysis is carried out. This analysis is used to assess the consistency of results across specific items (i.e., dimensions for this study) within a test. It is used in this study to indicate how closely the dimensions (F1, F2 and F3) are as a group in relation to the three tools tested. The most common measure of reliability of scores for a sample is the coefficient of internal consistency, or Cronbach's alpha (Cronbach, 1951). A Cronbach's alpha of 0.70 or higher is generally considered as an "acceptable" value of internal consistency. For example, according to Table 4, usefulness (F1) is measured using three detailed indicators: (D1) *trade-offs,* (D2) *swaps* and (D3) *conflict*. Looking at Table 6, the coefficients for F1 (0.838, 0.805, 0.845) indicate high internal consistency, which guarantees that the three indicators are a reliable measure of perceived usefulness. Similarly, reliability indicators are calculated for F3. Acceptable coefficients of internal consistency make it meaningful to compute a general value of perceived effectiveness for each tool (see bottom row of Table 6).

To compare the means of the measures obtained with each tool, paired samples *t*-tests were carried out. This test is recommended when different treatments are applied to the same participants and is also considered to be suitable for small sample sizes in similar studies (e.g., Inman et al., 2011). In this case, it is tested whether the difference between the measures obtained from two tools is different from zero. Since the traditional confidence interval for a mean difference is 95%, a significance value smaller than 0.05 indicates evidence that there is a difference in means across the paired observations. By running paired *t*-tests within the set of three tools, it is possible to establish whether the use of one tool leads to statistically different observations than another. For instance, it can be concluded that tool 1 is statistically different from the rest if two paired *t*-tests indicate that tool 1 is statistically different from tools 2 and 3. Table 7 and Table 8 summarize the results of the comparison between perceived and observed effectiveness as indicated in the frameworks presented, respectively, in Table 4 and Table 5.

Table 4
Framework with dimensions, indicators and controls of effectiveness for individual decision-making.

Dimension	Indicator of perceived effectiveness	Control for observed effectiveness
F1. Usefulness of information	I1. Usefulness of tool	C1. Individual performance
	D1. Usefulness to identify trade-offs	D1. Performance trade-offs
	D2. Usefulness to support swaps	D2. Performance swaps
	D3. Usefulness to address conflict	D3. Performance conflict
F2. Clarity of information	I1. Clarity of tool information	C1. Test of subjects' understanding
F3. Impact on decision process	I1. Valuation of tool	C1. Individual MCA Score
	I2. Influence of tool on decisions	C2. Parcels with a new land use
	I3. Performance of plan	

Table 5
Framework with dimensions, indicators and controls of effectiveness for group decision-making.

Dimension	Indicator of perceived effectiveness	Control for observed effectiveness
F1. Usefulness of information	I1. Usefulness of tool	C1. Group MCA Score
F2. Clarity of information	I1. Clarity of tool information	C1. Test of subjects' understanding
F3. Impact on decision process	I1. Valuation of tool	C1. Group MCA Score
	I2. Influence of tool on decisions	C2-a. Intensity of tool use
		C2-b. Time using tool
		C2-c. Group strategy
		C2-d. Performance conflict
	I3. Performance of plan	C3-a. Group MCA score
		C3-b. Parcels with new land use
		C3-c. Spatial pattern of land use map (clustering)

Table 6
Reliability (Cronbach's alpha) of indicators of detailed perceived effectiveness for each tool.

	Multiple-map color	Single-map quantitative	Single-map qualitative
F1. Usefulness	0.838	0.805	0.845
F2. Clarity	–	–	–
F3. Impact	0.693	0.750	0.802
Overall F1, F2, F3 (alpha)	0.668	0.731	0.516

Table 7
Comparison of average tool effectiveness for individuals.[a] A figure in bold denotes maximum, in italics denotes minimum and an asterisk (*) denotes a significant difference from the nearest neighbor.

Dimension	Indicator	Perceived effectiveness			Indicator	Observed effectiveness		
		Multi-map colors	Single-map quantitative	Single-map qualitative		Multi-map colors	Single-map quantitative	Single-map qualitative
F1	I1	2.267	*1.60	**2.483**				
Usefulness	D1	**1.866**	*1.413	1.767	D1	0.333	***0.700**	*0.259*
	D2	2.000	*1.345	**2.067**	D2	*0.400*	*0.400*	***0.783**
	D3	1.900	*1.036	**1.931**	D3	***0.800**	*0.400*	*0.400*
F2	I1	2.067	*1.700	**2.267**	C1	*0.367	**0.800**	0.700
Clarity								
F3	I1	*0.714*	*0.714*	***1.563**	C1	*1.568	4.490	***5.325**
Impact	I2	1.593	*1.519*	***2.407**	C2	***57.33**	*40.30*	44.20
	I3	*0.036*	0.179	***0.786**				
Overall		1.700	*1.378*	***2.119**				

[a] For details on indicators, refer to Table 4.

Bottom row of Table 7 shows that the single-map qualitative tool has the highest average overall perceived effectiveness, which is significantly different to that of the other two tools. The single-map quantitative tool shows the lowest overall perceived effectiveness. In particular, the quantitative tool has the lowest perceived effectiveness for usefulness and clarity (F1, F2), both with a significant difference from the other two tools. Based on observed effectiveness, the single-map qualitative tool performs slightly better than the single-map quantitative tool, which in turn clearly performs better than the multiple-map colors tool.

5.2. Relationships between perceptions and observations

This section further analyzes the differences between the various tools by comparing the indicators in Table 7 and Table 8. The findings of this analysis are organized in terms of the hypotheses posted in Section 3.3. More specifically, this section compares the average levels of effectiveness for the three tools with a focus on:

1) The extent to which the overall perceptions on the tool effectiveness run parallel to the MCA scores achieved at group level
2) The relationship between the amount of information supplied and the MCA scores of groups
3) The relationship between intensity of tool use and the final MCA score
4) The relationship between the performance of individuals and the MCA score of a group

5.2.1. What is the relationship between the perceptions of tool effectiveness and the MCA scores achieved on a group level?

This question tries to find out to what extent the perceptions can match the results attained. Indicators to address this question are: overall individual perception of tool effectiveness (Table 7, last row); and group MCA score (Table 8, F3, C1). Overall, the participants found the single-map qualitative tool to be the best, which differed significantly from the other two tools. As Table 7 shows, this result is also consistent on usefulness, clarity and impact, and clearly shows that the single-map qualitative tool was found to be the most effective. These results mean that the perceived degree of tool effectiveness runs parallel to the Total group MCA scores achieved using the tools.

5.2.2. What is the relationship between the amount of information supplied and the MCA scores of the plans generated with the help of tools?

The amount of information users are able to handle is a relevant aspect of spatial decision-making. The associated cognitive effort, which is the mental effort that people use for processing information, can also be a relevant aspect. Indicators of observed Total MCA score achieved by individuals (Table 7, F3, C1) and by the groups (Table 8, F3, C1) are used to address this question. Individual MCA scores show that the single-map qualitative tool helped participants produce the best plans. The use of the multiple-map color tool resulted in the lowest scores. Group MCA scores show that the qualitative tool helped the groups produce the best plans.

Table 8
Comparison of average tool effectiveness for groups.[a] A figure in bold denotes maximum, in italics denotes minimum and an asterisk (*) denotes a significant difference from the nearest neighbor.

Dimension	Indicator	Perceived effectiveness			Indicator	Observed effectiveness		
		Multi-map colors	Single-map quantitative	Single-map qualitative		Multi-map colors	Single-map quantitative	Single-map qualitative
F1	I1	2.267	*1.60	**2.483**	C1	6.098	*5.122*	***7.317**
Usefulness								
F2	I1	2.067	*1.700	**2.267**	C1	*0.367	**0.800**	0.700
Clarity								
F3	I1	*0.714*	*0.714*	***1.563**	C1	6.098	*5.122*	***7.317**
Impact	I2	1.593	*1.519*	***2.407**	C2-a	11.80	*9.200*	**13.25**
					C2-b	***04:07**	*02:51*	03:15
					C2-c	**0.680**	0.634	*0.612*
					C2-d	**0.800**	*0.400*	*0.400*
	I3	*0.036*	0.179	***0.786**	C3-a	6.098	*5.122*	***7.317**
					C3-b	**303.1**	*202.0*	261.0
					C3-c	**0.308**	*0.272*	0.289

[a] For details on indicators, refer to Table 5.

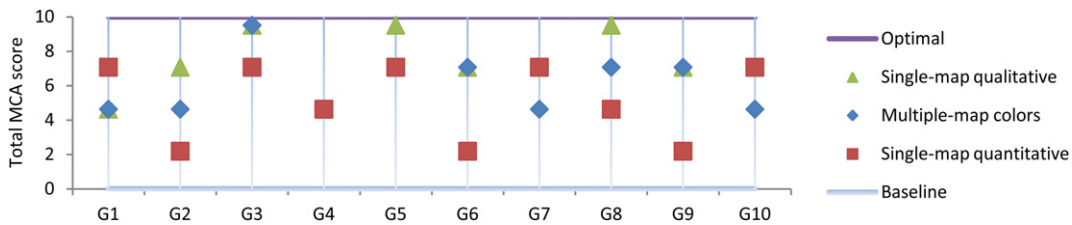

Fig. 8. Observed group effectiveness: Total MCA score for all groups.

The worst group plans were obtained with the single-map quantitative tool. Apparently, the optimal SDSS configuration takes place if it provides information in a qualitative way. This may reflect that the use of the single-map quantitative tool requires a high cognitive effort that is beyond the participant's capacity. It may also be that users of the information systems are just more competent and effective when using qualitative data instead of quantitative data. Anyhow, irrespective of the underlying mechanism, it can be concluded that more precise information is not necessarily better than qualitative information and that even the opposite may occur.

5.2.3. Does the degree of discussion and cooperation triggered by the use of the tools have an impact on the group MCA scores?

When comparing the ranking of the three tools according to the group MCA scores (Table 8, F3, C1) with their performance in terms of their intensity of use of tool information, total time use and the extent to which they trigger collaborative strategies (F3, C2-a–C2-c), it is found in the first place that the three methods do not differ significantly in terms of the extent to which they trigger intensive group interactions, so this is not a strong factor to explain differences in MCA group performance. The total time use involved with the three tools is different, but the rank correlation with the group MCA score is low. The clearest relationship is found for intensity of use of tool information. The success of the single-map qualitative tool may well be due to the fact that it is in such a format that it stimulates group members to use the tool information in a large number of rounds.

5.2.4. What is the relationship between the performance of individuals to carry out specific land use allocation tasks and the total group MCA score generated with the help of the tools?

This question relates to how the correct use of a tool affects the achieved results. To address this question, the following indicators are used: observed individual performance in the tasks 'identification of trade-offs', 'completion of land use swaps' and 'addressing conflicts in objectives' (Table 7, F1, C1, D1-D2-D3) and both individual and Total group MCA scores (Table 7, F3, C1 and Table 8, F3, C1, respectively). According to observed effectiveness, 70% of the participants used the single-map quantitative tool correctly to identify trade-offs. 97% of all participants used the qualitative tool correctly to perform land use exchanges. The multiple-map color tool proved to be the best in helping participants address conflicts of objectives. Group MCA scores indicate that the use of the qualitative tool resulted in the best plans. These results suggest that good individual performance in land use allocation tasks leads to plans with high MCA scores. The qualitative tool allowed for the best individual performance in executing swaps. The multiple-map color tool was the best to address conflict of objectives. Group MCA scores show that the single-map qualitative tool allowed for the best plans. It can be concluded that a good individual performance in spatial decision-making contributes to some extent to an increase in the Total group MCA score. Fig. 8 shows that all groups succeeded in increasing the baseline Total score. It is also clear that the single-map quantitative tool provided the worst group results. Therefore, the qualitative tool was the most successful in increasing the Total score on both individual and group levels.

5.3. Spatial effects of support tools

The above discussion focused on the quality of the changes in land use as measured by means of the overall indicator based on MCA. The actual land use changes and spatial patterns as they can be observed on a map have not been addressed thus far. This section discusses explicitly these dimensions, particularly how these changes relate to collaborative work. It looks particularly at indicators 'parcels with a new land use' (Table 8, F3, C3-b) and Moran's value for spatial patterns of land use (Table 8, F3, C3-c). Each group generated three land use maps, thus a total of 30 maps were generated by all groups.

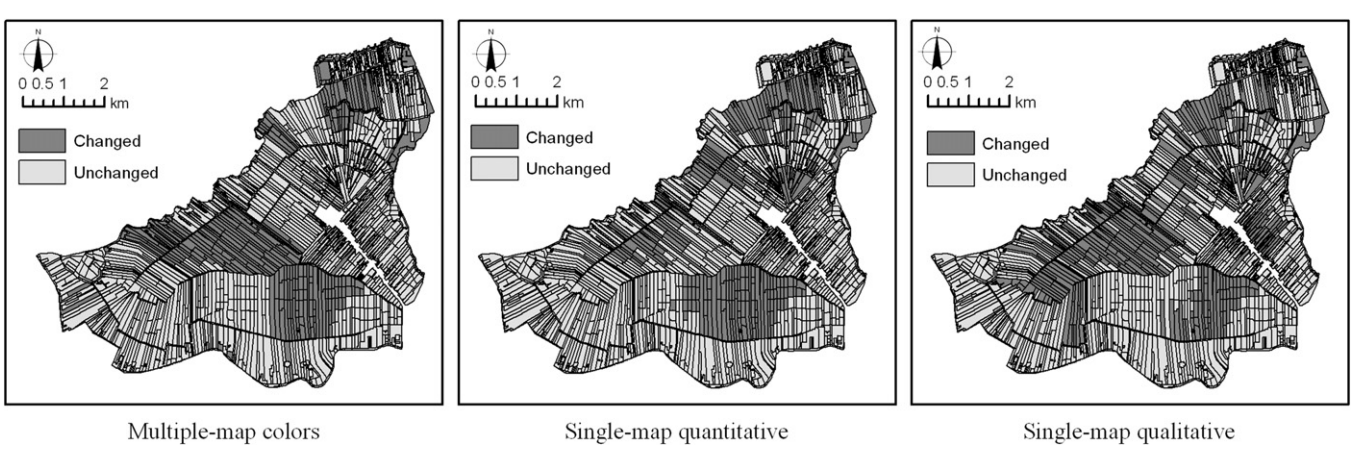

Fig. 9. Changes in land use with respect to the reference situation for the three plans generated by group 5.

Table 9
Average number of parcels with different land use. A figure in bold denotes maximum, in italics denotes minimum and an asterisk (*) a significant difference. Figure in brackets corresponds to percentage of all parcels.

	Reference	Multiple-map colors	Single-map quantitative	Single-map qualitative
Reference	–	**303.1 (21%)**	**202.0 (21%)*	261.0 (18%)
Multiple-map color	303.1 (21%)	–	**303.7 (21%)**	*300.1 (20%)*
Single-map quantitative	**202.0 (14%)*	**303.7 (21%)**	–	281.2 (19%)
Single-map qualitative	*261.0 (18%)*	**300.1 (20%)**	281.2 (19%)	–

Table 10
Distribution of probability that a parcel's land use is changed as the result of group decisions with each of the three tools. Figure in brackets corresponds to percentage of all parcels.

	Multiple-map colors	Single-map quantitative	Single-map qualitative
0–20%	860 (58%)	1013 (69%)	976 (66%)
20–40%	333 (23%)	351 (24%)	275 (19%)
40–60%	232 (16%)	89 (6%)	127 (9%)
80–100%	53 (4%)	25 (2%)	100 (7%)

5.3.1. Difference maps

To compare how different the maps are between the three tools and with respect to the reference land use situation, a difference map between two situations is created. A difference map shows whether or not a change in land use took place on a parcel after the use of a tool. Therefore, the legend of a difference map includes two non-overlapping classes: *unchanged* land use and *changed* land use. Fig. 9 exemplifies the difference maps for the plans generated by Group 5. Parcels that have a changed land use with respect to the reference map are shaded dark. The map made with the single-map qualitative tool shows the most apparent differences. The same procedure was followed to create difference maps for all groups. Difference maps are important for two reasons. First, they focus on a very relevant aspect of land use planning: policy alternatives are not only represented in terms of their scores in 'objective space' where MCA scores are compared, but also in the physical space where the intervention in terms of proposed land use changes are represented. They are thus of direct relevance for participants in the process since they inform them on where the proposed land use changes will take place. The second reason is that by comparing them across various decision support tools, it becomes possible to analyze to what extent a certain tool is stimulating a more explorative attitude than others. This is important since one of the functions of SDSS is to stimulate and support creativity among participants in the decision-making process. A tool that leads to outcomes that are far away from the reference situation may be a sign that it supports a search outside the conventional domain.

Table 9 summarizes the average number of parcels whose land use got changed as a result of the group assignment carried out by each of the 10 groups. The study area comprises a total of 1478 parcels. Statistically, just as in the individual assignment, the maps produced by the groups using the multiple-map color tool present the largest average of parcels with a modified land use with respect to the reference situation (as can be seen from Table 2, F3, C2 and Table 3, F3, C3-b). However, this result shows no significant difference to that of the qualitative tool. On average, the quantitative tool shows the least amount of unmodified parcels, again just as in the individual assignment. This result also matches observed results of group performance, namely strategic information use, time used for discussion and information use, and extent of information use to address conflicts.

To compare the spatial results of each tool with each other, a similar analysis was conducted, having each tool as the reference. This analysis reveals no statistical difference among each possible pair of tools, meaning that there are no signs of learning in the sense that the groups would base their land use plans on the outcomes of earlier sessions. Fig. 10 shows three maps that illustrate the percentage of land use change with respect to the reference plan. Table 10 summarizes these results. The bottom row of this table shows that the single-map qualitative tool has the largest number of parcels (100) whose land use was changed in more than 80% of all plans made by the groups, whereas the single-map quantitative tool allowed for the smallest number of parcels. This ranking matches those of Total group MCA score and observed intensity of tool use. The differences in tool performance are indeed reflected by the volume of proposed changes in land use. It appears that the single-map qualitative tool is most effective in exploring alternatives that are rather different from the reference situation. Although it is certainly not necessarily true that thinking 'out of the box' leads to better results, the present study finds that the two dimensions tend to be in harmony.

5.3.2. Differences in spatial land use patterns

The spatial patterns of the maps produced by all groups with the three tools were compared with a spatial autocorrelation analysis. This analysis measures parcel similarity for one land use map based

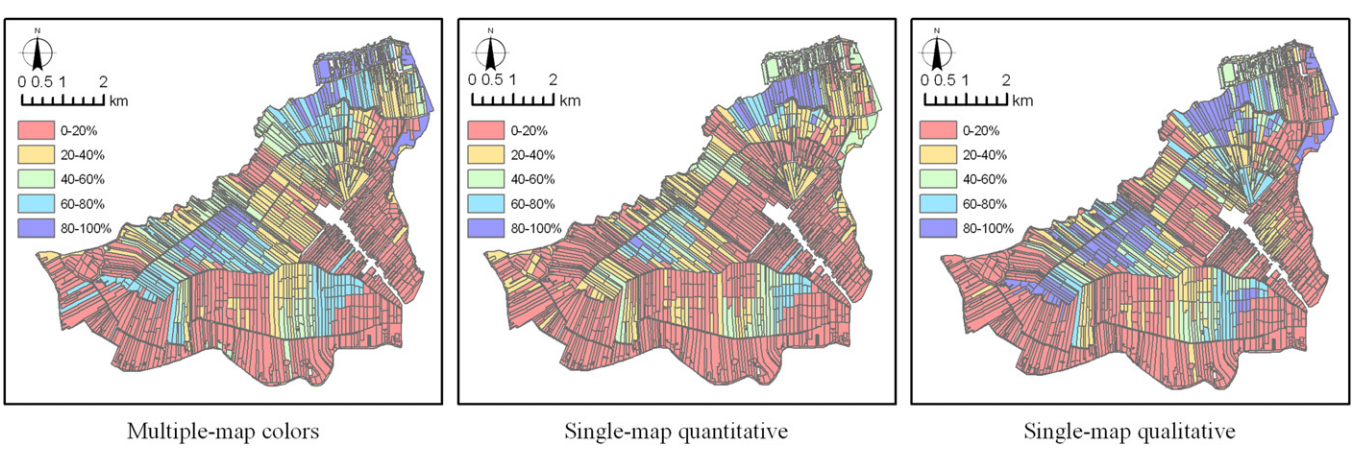

Fig. 10. Probability that a parcel's land use is changed as the result of group decisions with each of the three tools. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

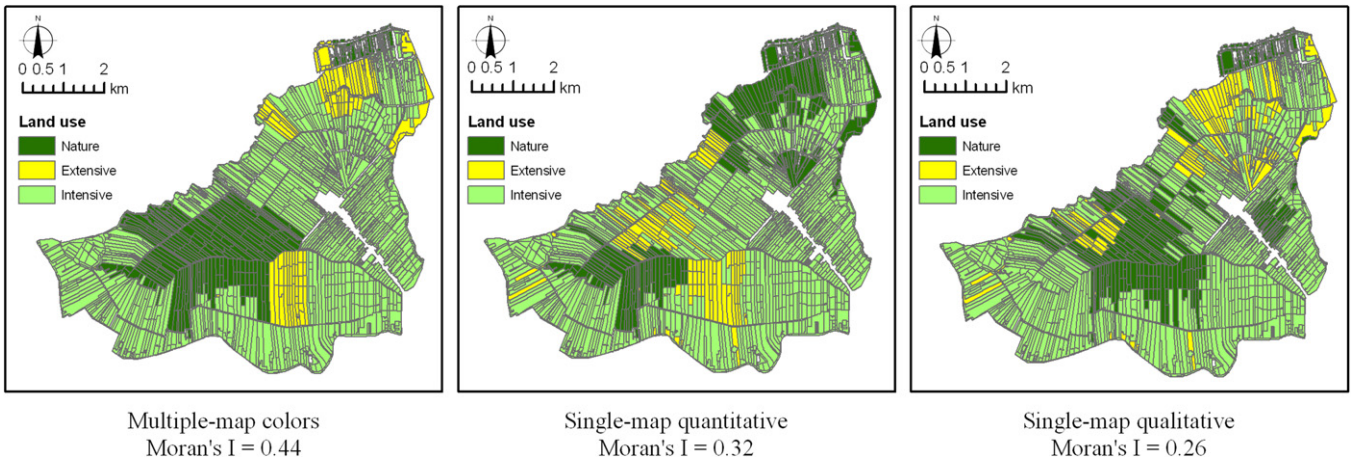

Fig. 11. Difference in spatial pattern for the three plans generated by group 5. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

on parcel locations and land use types, evaluating whether the pattern is clustered, dispersed, or random. Spatial autocorrelation can be estimated with the Moran's I index (Moran, 1950). A Moran's I value near +1 indicates clustering, while a value near -1 indicates dispersion. Fig. 11 shows the three maps generated by Group 5, each with its corresponding Moran's I value.

By visual inspection, it becomes apparent that the map produced with the multiple-map color tool presents the most clustered land use patterns of the three maps, which is supported by looking at its Moran's I value. Moran's indexes were calculated for all 30 maps. Statistically, the maps produced on the basis of the multiple-map color tool present the largest clustering tendency (see Table 8, F3, C3-c). This result is in agreement with that for spatial differences (see Section 5.3.1) as it matches observed group performance measures, namely land use change, strategic information use, time used for discussion and information use, and extent of information use to address conflicts. This suggests that the multiple-map color tool, which provides feedback as colors on printed maps, leads to plans of higher spatial consistency that those attained with the other two tools, which provide more detailed feedback on individual parcels.

6. Conclusions

MCA and GIS are promising tools for the translation of output of integrated models into integrated knowledge for policy makers. They may help to reduce the gap between the worlds of science and policy making. These tools can be combined in spatial decision support tools for collaborative decision-making, but the research assessing the effectiveness of these decision support tools is still ongoing. To date, there are no concrete guidelines in the literature about the evaluation of tool effectiveness. Previous empirical studies have typically reported on qualitative assessments, focusing on positive perceptions of end-users as a measure of success. This study takes a step forward by evaluating the effectiveness of three collaborative map-based tools on the basis of both user perceptions and objective measurements of performance in the execution of specific tasks. This evaluation is based on a framework that included tests of usefulness of tools, clarity of information and impact on decisions. In drawing conclusions, this section discusses the results on the basis of three aspects: first, the comparison between the tools tested; second, the comparison between individual and group decision-making; and third, the implications for integrated environmental assessment and policy making.

6.1. Effectiveness of map-based decision support tools

This section addresses the first three hypotheses posted in this study. The first hypothesis was: "*the perceptions of tool effectiveness run parallel to the performance outcomes on a group level*". Testing this hypothesis involved comparing the perceived effectiveness of each tool with observed group measurements. Of the three tools tested, the participants found the single-map qualitative tool to be the most useful, followed by the multiple-map color tool. These judgments largely match observations of individual performance: first, the use of the qualitative tool prompted the best individual performance in decision-making tasks linked to land use allocation, such as the identification of potentially favorable trade-offs on the map and the resulting reallocation of land use between the parcels in question; second, the use of the qualitative tool also resulted in the best plans on both an individual and a group level.

The second hypothesis, "*the amount of information supplied in the tools affects the participants' understanding of the tools*", relates to the clarity of the information presented in the tools. This hypothesis follows up on recommendations by previous studies (e.g., Kennedy and Bishop, 2008) that further controlled experiments can contribute to measuring how much knowledge of choices of participants using tools influences choice behavior. It was found that the tool with the highest information level, that is, the single-map quantitative tool, proved to be too difficult for the participants to use and was also considered by them to be the least useful. Using the quantitative tool thus resulted in the worst plans for both individuals and groups, despite a high participants' understanding of this tool.

For the third hypothesis "*the intensity of tool use affects the performance outcomes on a group level*", it was found that during the group assignment the qualitative tool was used the most extensively to support decisions. The use of the multiple-map color tool took the most time and prompted the longest discussions. Footage showed that when conflicts emerged during the negotiations, this tool proved to be the most used tool to address such conflicts. Overall, these findings suggest that presenting the same results of spatial MCA in different ways to support negotiated land use allocation leads to different outcomes, that is, land use plans. In this study, the use of qualitative feedback offered on parcels on one digital map led to the best land use plans both individually and collectively. Furthermore, the use of the multiple-map color tool also allowed for the spatially most different plans both on an individual and group level, and for the spatially most autocorrelated plans on a group level. Testing hypotheses 2 and 3 shows that, in this application, a low level of complexity in the information with which decision-makers are

provided prompted extensive allocation work and led to overall results of higher quality. Previous studies reported different success rates of tools for different complexity levels of information supplied (e.g., Nyerges et al., 2006; Dias, 2007).

6.2. Individual vs. group effectiveness

Measuring the extent to which the performance of individuals in using decision support tools influences group outcomes has not been explicitly studied in the literature. Most studies discussed in this article have only utilized individual perceptions of effectiveness in their analyses (e.g., Salter et al., 2009; Nyerges et al., 2006; Inman et al., 2011). The fourth hypothesis of this study, "*the individual performance in using the tools influences the outcomes on a group level*" addresses this issue. The average performance of individuals to achieve high-quality plans, that is, the Total individual MCA score, matches that achieved on a group level. With the single-map qualitative tool, the highest MCA scores were achieved both individually and collectively, and with a significant difference from the other two tools. In terms of observed individual task performance, it was found that the specific individual ability of the participants to identify and use potentially favorable trade-offs on the map has an effect on the quality of the results achieved as a group. In terms of group performance, some results correlated significantly with the Total group score. In particular, the extent of discussion among participants triggered by the use of the support tools proved to have both a positive and significant impact on the quality of group results. It can be concluded that irrespective of the type of information supplied to users, in this study both the individual ability to use this information to support negotiated exchanges on land use and the extent of the discussion triggered have an impact on the quality of the resulting land use plans.

Regarding the spatial configuration of the generated land use plans, the difference maps also showed a full correspondence between individual and group performance. In particular, the extent to which the spatial patterns of the reference situation were modified correlates with the sequence of tool exposure: the participants changed the land use of the largest amount of parcels on both an individual and a group level using the multiple-map color tool and the least amount with the single-map quantitative tool. This result also matched the individual MCA scores as well as group strategic information use, time used for both discussion and information use, and information use for addressing conflicts. It can be concluded that the individual performance to generate high-quality plans has an impact on the extent of discussion and strategic information use, which is also reflected on the relative spatial changes of the resulting plans.

6.3. Implications for integrated environmental assessment and policy making

On the basis of this study's specific experiment, a number of conclusions can be formulated that apply in a more general setting of collaborative SDSS. It was found that MCA and GIS are effective tools to provide intelligible information arising from an integrated model for policy contexts. The present experiment basically compared a number of tools involving different degrees of detail and ways of spatial representation of the integrated information. A major conclusion is that the cognitive effort required to understand different levels of information content is a serious issue in spatial decision support. The experiment showed that cognitive effort can in fact change with different approaches for presenting output information of spatial decision support tools. This holds true for both the amount of information, the level of detail of information to be supplied to end-users and also for the format in which the information is provided. The single-map quantitative tool, being strong in detailed information on trade-offs between policy objectives, performed clearly worse than the multiple-map color and the single-map qualitative tool. Further empirical studies focusing on spatial tool development should explore the effects in terms of cognitive effort of the amount of information and detail provided in a land use planning support tool. The survey-based methodology of a study by Grotenhuis et al. (2007), which investigated how to reduce cognitive effort in travel information provision for public transport, can provide useful guidelines. Additionally, Nyerges et al. (2006) recommended software assistance by a facilitator-chauffeur to help reduce cognitive effort.

A second conclusion is that the perceived effectiveness of tools has a rather close relationship with observed effectiveness. The high-tech single-map quantitative tool was valued clearly lower than the low-tech multiple-map color tool. This means that the fear for risk of overestimation of the utility of high-tech decision support just because it gives the users a better feeling is not confirmed in this case. The conformity of this finding with the literature discussed in this article is not entirely clear. While Dias (2007) found that perceived added value increases when tool information becomes more detailed, Janssen and Uran (2003) concluded that a high preference for a decision support tool does not always relate to a successful use. Inman et al. (2011) reported that perceived effectiveness of tools is influenced by end-user's employment. Users of SDSS need training and time to play and get acquainted with the tools. The experiments showed that a considerable part of the participants involved did not fully grasp how the tools work. Nevertheless, all tools proved useful in terms of the improvement of the overall land use quality. The intensity of tool use is clearly correlated with the quality of the overall outcome. This result indicates that the various tools used were in fact useful for the group task: in the present experiment, SDSS do what they are supposed to do.

Furthermore, it appears that when a tool is more stimulating, it helps to achieve more effective use of information. Support tools appear to be rather different in terms of the extent to which the proposed alternative is far away from the reference situation. It appears that the tool based on qualitative information leads to alternatives that are further away from the reference situation compared with the other tools. Further research on how properties of tools trigger exploratory attitudes among participants in land use planning is recommended. The planning process, for which the tools were made, includes experts from many disciplines and backgrounds. Bringing these experts into one workshop could be called a form of integrated environmental assessment. In the workshop, information from all sources was presented as maps on the table. Using maps as the single means of communication and a common assignment on the 'Touch table' to stimulate cooperation proved to be a very effective means of communication between stakeholders with different backgrounds.

Acknowledgments

This research was funded by the Dutch National Research Programs Climate changes Spatial Planning (project ME6) and Living with Water (project 'Waarheen met het veen?'). The authors would also like to thank Juan Pablo Mendoza for his sound statistical advice and Johanna Castro Mendoza for her lab assistance during the experiment.

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